Term Test 1 Solutions 2021

Calculus (University of Toronto)



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University of Toronto
Faculty of Arts and Science

MAT137Y Test 1

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Duration: 110 minutes

No Aids Permitted

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1. For each of the following questions, only your final answer will be graded.

No justification is necessary.

(1a) (2 points) Let $L \in \mathbb{R}$ and f be a function defined on an open interval of the form (p, ∞) for some $p \in \mathbb{R}$. Give the complete formal definition for $\lim_{x \to \infty} f(x) = L$.



(1b) (2 points) Let $a \in \mathbb{R}$ and f be a function defined on an open interval centered at a, except possibly at a. Give the complete formal definition for $\lim_{x \to a} f(x)$ does not exist.



For (1c) to (1f) below, evaluate the limit. If the limit does not exist and it's not $\pm\infty$, put "DNE" in the box.

(1c) (1 point) $\lim_{y \to 0} \left\{ \frac{3}{y} - \frac{3}{y^2 + y} \right\}$

Final Answer



(1d) (1 point) $\lim_{t \to 0} \frac{\sin(4t)\cos(t^2)}{\sin(7t)}$

Final Answer



(1e) (1 point) $\lim_{x \to -\infty} \frac{4x^2 + \sqrt{4x^6 + x}}{2x^3 - 2}$

Final Answer



(1f) (1 point) $\lim_{\theta \to \infty} \frac{\sin\theta}{\theta}$

Final Answer



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2. For each of the following questions, only your final answer will be graded.

No justification is necessary.

(2a) (1 point) Let f be a function defined on $\mathbb{R}$. Assume we know that

$$\forall x \in \mathbb{R}, 0 < |x-4| < 0.2 \implies |f(x)-7| < 0.4$$

If the limit $L = \lim_{x \to 4} f(x)$ exists then what are the possible values of L?

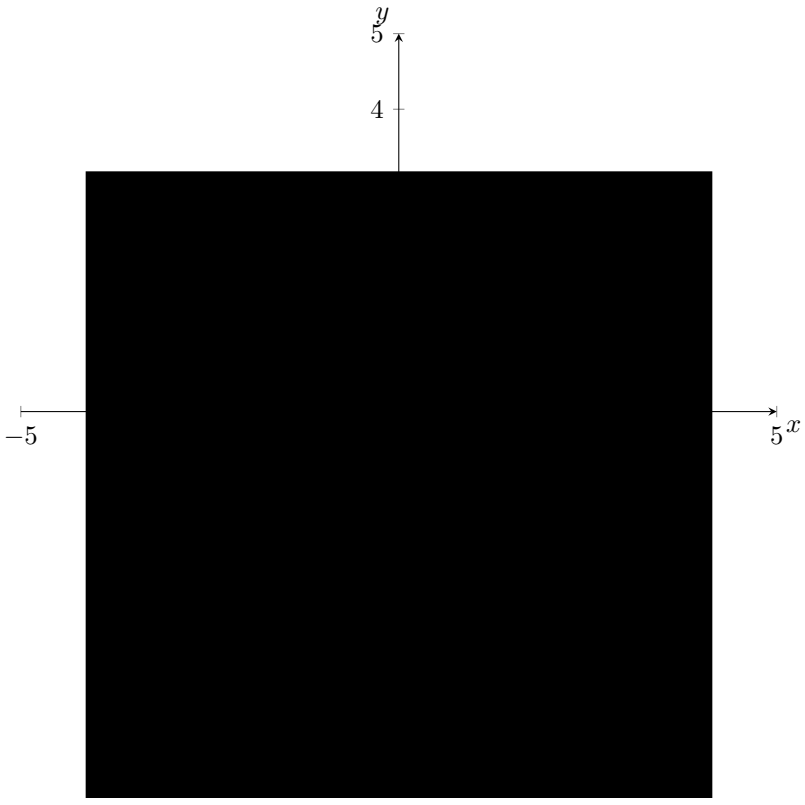
Final Answer

[redacted]

(2b) (5 points) Sketch the graph of a function g defined on $(-5, 5)$which satisfies the following properties at once:

- $\lim_{x \to 2} g(x) = 3$
- $\lim_{x \to 2} g(g(x)) = -1$
- $\lim_{x \to -2} g(x)$ does not exist
- $\lim_{x \to -2} [g(x)]^2$ exists
- g has a removable discontinuity at $x = 0$.

[redacted]



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3. For each of the following questions, only your final answer will be graded.

No justification is necessary.

(3a) (1 point) Write down the set of positive odd integers using set building notation.

Final Answer

[redacted]

(3b) (1 point) Write down the sentence which is the negation of the following statement:

On every page of the phone book, there is a phone number whose first digit is 2.

Final Answer

[redacted]

(3c) (6 points) For each statement, determine whether it is TRUE or FALSE. Circle the correct answer. Again, no justification is necessary. Here, we define $\mathbb{N} = \{0, 1, 2, 3, \cdots\}$.

i) $\forall x, y \in \mathbb{Z}, \exists z \in \mathbb{N}$ s.t. $x + y < z$ TRUE FALSE

ii) $\exists x, y \in \mathbb{Z}$ s.t.$\forall z \in \mathbb{N}$, $x + y < z$ TRUE FALSE

iii)IF $|f|$ is continuous at a THEN f is continuous at a. TRUE FALSE

iv)IF there exists $x \in \mathbb{R}$ s.t.$x^2 < 0$ THEN $3 < 2$. TRUE FALSE

v) Let f and g be functions defined on $\mathbb{R}$ and let $a \in \mathbb{R}$. TRUE FALSE
IF $\lim_{x \to a} f(x) = 0$ THEN $\lim_{x \to a} f(x)g(x) = 0$

vi) Let f, g, and h be functions defined on $\mathbb{R}$ and let $a \in \mathbb{R}$. TRUE FALSE
IF

- $\forall x \in \mathbb{R}$, $f(x) < g(x) < h(x)$
- $\lim_{x \to a} f(x)$ exists and $\lim_{x \to a} h(x) = \infty$

THEN $\lim_{x \to a} g(x)$ exists or is ∞.

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4. (4a) (2 points) Let $a, L \in \mathbb{R}$. Let f be a function defined on $\mathbb{R}$. Write the formal definition of $\lim_{x \to a} f(x) = L$.

[redacted]

(4b) (4 points) Use the formal ϵ-δ definition of the limit to prove

$$\lim_{x \to 1} \frac{3x+2}{2x+3} = 1$$

Do not use any limit laws or any other theorems.

[redacted]

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5. (6 points) Let f be a function defined on an interval centered at 4.

(a) (2 points) Write the formal ϵ-δ definition of " f is continuous at 4".



(b) (4 points) Assume f is continuous at 4 and $f(4) > 0$. Prove that there exists an open interval I, centered at 4, such that

$$\forall x \in I, f(x) > \frac{f(4)}{2}$$

Write a proof directly from the above formal ϵ-δ definition of continuity. Do not use any of the limit laws or any other theorems.



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6. Let f and g be two strictly positive functions with domain $[1, \infty)$. We say g *controls* f if

$$\exists C > 0 \text{ s.t. } \forall a, b \in [1, \infty), |f(a) - f(b)| \leq C|g(a) - g(b)|$$

Define functions A and B by $A(x) = 3x$ and $B(x) = x^2$ for all $x \in [1, \infty)$.

(6a) (3 points) Show that B controls A.

[redacted]

(6b) (3 points) Show that A does NOT control B.

[redacted]

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